

A GIFFEN EXAMPLE

ECON 101–INTRODUCTION TO MICROECONOMICS

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ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

This problem is inspired by the paper Haagsma [2012]. I originally designed this problem for an Intermediate Microeconomics class I taught back in the US while I was a graduate student. It does require some knowledge of multivariate calculus to comprehend fully, so you may eventually want to revisit this further down the road.

This problem is just for fun, just take a look if you have time and are interested. However, you can still play around a little bit with the things that you know from my ECON101 class: try graphing the demand for good x_1 below using the parameters I give in item 7). You will see that it does exhibit Giffen Behaviour!

A special case of the utility function defined in the paper is as follows:

$$u(x_1, x_2) = \ln(x_1 - 1) - 2 \ln(5 - x_2).$$

Suppose x_1 is consumption of food and x_2 is consumption of alcohol. The utility function is defined for $x_1 > 1$ and $0 \leq x_2 < 5$. The interpretation of this utility function is that the consumer needs to consume at least 1 unit of food to subsist and that consumption of alcohol is constrained to be less than 5 units, say because of the risk of alcohol intoxication. Let p_1 be the price of food and p_2 be the price of alcohol, m denotes the Income of the consumer.

1. Show that the consumer has positive marginal utility for both goods and therefore, will satisfy his budget constraint with equality. In particular, note that the marginal utility of alcohol increases the more of it is consumed, representing the *addictive behavior* of this good.
2. State this consumer's utility maximization problem. You can try to solve it if you are courageous. At an interior optimum, the optimal solutions (x_1^*, x_2^*) must satisfy $x_1^* > 1$ and $0 < x_2^* < 5$.
3. Irrespective of whether you were able to solve the consumer's problem or not, here I provide the demands that come from an *interior solution*:

$$\begin{aligned} x_1^* &= 1 - \frac{m - p_1 - 5p_2}{p_1}, \\ x_2^* &= 5 + \frac{2(m - p_1 - 5p_2)}{p_2}, \end{aligned}$$

assuming

$$m - 5p_2 < p_1 < m - 5p_2 + \frac{5p_2}{2} \quad (\star)$$

Why do we need the condition $(\star)$?

4. Show that food is an *inferior good* by computing $\frac{\partial x_1^*}{\partial m}$ and determining its sign.
5. Compute $\frac{\partial x_1^*}{\partial p_1}$. What is its sign? Under what condition on prices and income does the ordinary demand curve for food is *positively sloped*?
6. Show that alcohol is a complementary good to food by computing $\frac{\partial x_2^*}{\partial p_1}$ and determining its sign.

7. Let $m = 10$ and the price of alcohol be $p_2 = 1$. Suppose that we start at a price for good of $p_1 = 7$ and then it decreases to $p_1 = 6$. Compute the *substitution*, *income* and *total effects* of such a price decrease. What sign is the substitution effect? What sign is the income effect? What sign is the total effect? Does this corroborate the results from the previous items?
8. Putting everything together would yield the following explanation of why the demand for food exhibits Giffen behavior: When the income level is high enough, as is also the pleasure from hazardous drinking, a price fall of food may cause an income effect that stimulates hazardous drinking so much that this crowds out consumption of food. Can you think of other examples in which this explanation makes sense? Provide some other examples.

References

Rein Haagsma. A convenient utility function with giffen behaviour. *International Scholarly Research Notices*, 2012.